P-ADON4PDF: Prior-Enhanced Attention Document OCR Network for Image-to-Structure

Minghao Li 2330034027
Ruiyan Yin 2330034066
Xuhao Zhang 2330034073

Group 27

https://github.com/SuleynanAuir/P-ADONIS

Abstract

*Scene text images in documents like raw scanned PDFs file often suffer from low resolution and noise, hindering accurate text recognition. We propose **P-ADON4PDF**, a Text Prior-Enhanced and attention mechanisms with a diffusion model to improve the recognizability of text images in low-resolution (LR) with image-to-structure conversion. Our method integrates an attention-based module and a diffusion-based prior enhancement module to address two key factors affecting text recognition: both visual text structure and semantic information. Experiments on benchmark datasets (e.g., TextZoom [3]) demonstrate a good performance in both image quality and OCR accuracy.*

1. Introduction

Reading text from low-resolution document images is a significant challenge in Optical Character Recognition (OCR) due to issues like blur, noise, and small font sizes, which hinder detection accuracy. Scene Text Image Super-Resolution (STISR) aims to improve these images by enhancing visual sharpness while also making them more OCR-friendly [1]. Unlike general image super-resolution, STISR must preserve both the global text structure and semantic correctness [1,4]. Existing methods often fail to achieve both, leading to visually improved images that still face recognition issues, as seen in models like TATT and DPMN [1]. To address this, we introduce P-ADON4PDF, a model that enhances super-resolution by incorporating learned text priors. By combining attention-based feature modeling with diffusion-driven prior enhancement, our method targets both structural integrity and semantic fidelity. This results in super-resolved images that are clearer and more reliable for OCR. With this approach, we aim to set a new standard in STISR, improving both image clarity and OCR accuracy for low-resolution text recognition.

2. Method

Our group addressed the issues that scene text often has different lengths, deformations, and orientations, as well as the problem that existing text priors obtained from LR image recognition devices are of low quality, making them difficult to be strong guidance. We introduced an attention-based text prior generation module (TPGM) and a text prior enhancement module (TPEM) to better capture long-distance dependencies between characters and structure.

2.1. Goal

Design an end-to-end network that integrates attention mechanisms (to restore visual structure), diffusion-enhanced text priors (to improve semantic information), and multi-task learning (SR+recognition) to generate clear and readable SR images.

2.2. TPGM: Text Prior Generation Module

The Text Prior Generation Module (TPGM) involves several steps, including feature extraction, feature sequence encoding, and probability sequence generation.

Input:

- $\mathbf{I}_{lr}$: The original LR feature input. which is a 2D feature map extracted by Swin Transformer block.

Output:

- $\mathbf{P}_l$: The primary text prior, a sequence of probability distributions.

Step 1: Get Input ($\mathbf{I}_{lr}$)

- The input $\mathbf{I}_{lr}$ is a 2D feature map by a Swin Transformer feature extraction block. Then we get extracted feature map: $\mathbf{F} \in \mathbb{R}^{H' \times W' \times C}$

Step 2: Feature Sequence Encoding

- **Feature Compression:** The 2D feature map $\mathbf{F}$ is converted into a 1D sequence $\mathbf{S}$ for sequence prediction.

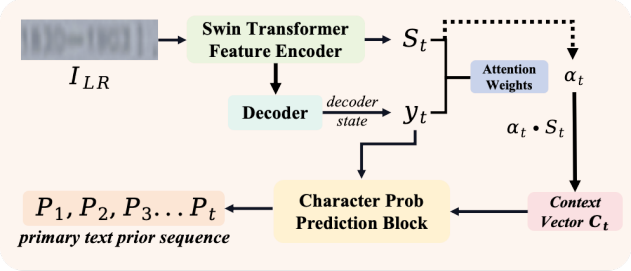


Figure 1. **Overview of Text-Prior Generation Module (TPGM):** This figure illustrates a Swin Transformer-based text generation model framework, which extracts features from an image and associates them with a text prior sequence to generate character-level predictions.

- **Processing:** To compress height dimension H' to 1, we perform *average pooling along rows height dimension* of the feature map. Then we get a compressed 1D feature sequence $\mathbf{S} \in \mathbb{R}^{W' \times C}$.

Step 3: Generate a Probability Sequence

- **Attention Weights:** At each step t, the decoder state $\mathbf{y}_{t-1}$ is compared with the encoder sequence $\mathbf{S}$. The attention weights α_t are calculated as:

$$\alpha_t = \text{Attention}(\mathbf{S}, \mathbf{y}_{t-1})$$

where $\alpha_t \in \mathbb{R}^{W'}$ is the weight vector indicating the importance of each position in the input sequence $\mathbf{S}$ when generating the current character.

- **Context Vector Generation:** The context vector $\mathbf{c}_t$ is generated based on the attention weights as follows:

$$\mathbf{c}_t = \sum_{j=1}^{W'} \alpha_{t,j} \cdot \mathbf{S}_j$$

- **Character Probability Prediction:** decoder combines the output of previous step $\mathbf{y}_{t-1}$ with the context vector $\mathbf{c}_t$ to predict character probability distribution at time step t, denoted as $\mathbf{p}_t$:

$$\mathbf{p}_t = \text{Softmax}(\text{Linear}([\mathbf{h}_t; \mathbf{c}_t]))$$

Step 4: Output Primary Text Prior ($\mathbf{P}_l$)

- We concatenating probability distributions $\mathbf{p}_t$ generated at all time steps t into the **primary text prior sequence** $\mathbf{P}_l$:

$$\mathbf{P}_l = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_L]$$

2.3. TPEM: Text Prior Enhancement Module

Motivation: We find some problems with traditional text prior extraction methods, and borrow the idea from diffusion framework to help with higher-quality text priors.

- We find that Traditional methods rely on text priors ($\mathbf{P}_l$) generated by the LR image recognizer, and this is heavily affected by image blurring.
- Using text priors ($\mathbf{TP}_{HR}$) generated from HR images is not available during testing on LR images.

Our team proposes TPEM to make the text prior $\mathbf{P}_l$ closer to $\mathbf{TP}_{HR}$ in quality. TPEM generates an text prior via a conditional diffusion process, inspired by recent diffusion-based text restoration models [2]. During training, diffusion module learns to produce a clean high-resolution text from a noisy conditioned on LR image y [2].

Implementation: Diffusion Framework (DDPM)

- **Forward Diffusion:**
  - Gradually add Gaussian noise to the HR text prior ($\mathbf{P}_h$), turning it into $\mathbf{x}_T$..

  Then we get a noisy version of the HR text prior with noise: HR prior + Noise
- **Reverse Diffusion:** For reverse diffusion, our goal is to predict an enhanced prior approximates HR prior quality, enabling better predictions for new data. Start from noise on LR prior ($\mathbf{P}_l$) to estimate the enhanced prior ($\mathbf{P}_e$) using a denoiser function f_θ.

$$\text{HR}_{prior} + LR_{prior} \xrightarrow{\text{denoiser}} \text{Enhance}(\mathbf{P}_e)$$

Loss Design: The model is trained in two phases: an optional prior pre-training and fine-tuning of the full model. During full model training, a multi-task loss is adopted:

$$L_{\text{total}} = L_{\text{SR}} + \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{prior}} L_{\text{prior}} + L_{\text{MAE}} + L_{\text{CTC}}$$

- L_{SR}: the image reconstruction loss (SSIM).
- L_{rec}: the text recognition loss computed by pre-trained recognizer (ASTER, CRNN, MORAN, PARSeq).
- L_{prior}: the diffusion prior loss (optional: during joint fine-tuning) [3, 5].
- L_{MAE}: the Mean Absolute Error (MAE) loss (P_e/P_h)
- L_{CTC}: the Connectionist Temporal Classification (CTC) loss.

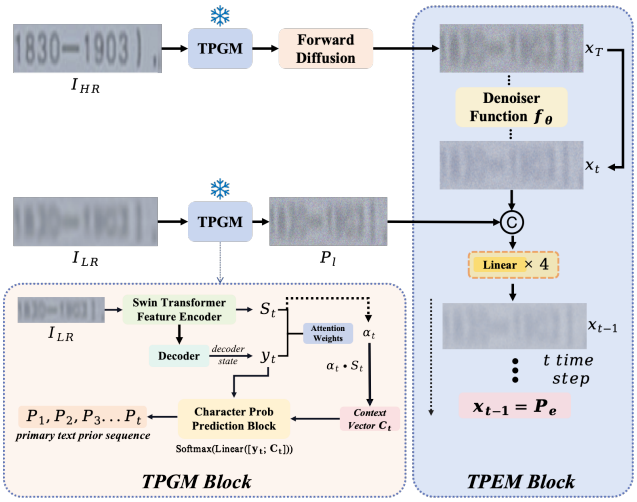


Figure 2. **Overview of Text-Prior Enhanced Module (TPEM):** illustrates the architecture of a model that integrates TPGM and TPEM blocks for feature extraction and diffusion-based denoising. The left side shows the TPGM block. On the right side, the TPEM block performs forward diffusion and denoising. It utilizes linear transformations to generate clean image x_{t-1} from noisy x_t, with each step informed by the context vector and attention weights.

2.4. AMM: Attention-based Modulation Module

Motivation:

- Traditional STISR methods often use CNN + BiLSTM structures like Sequential Residual Blocks, SRB.
- These methods are hard to capture long-range dependencies between characters, especially when text is long, deformed, or curved.

Our group make the Attention-based Modulation Module (AMM) consists of N modules, each made up of Convolutional layers, Local Attention Module (LAM), and Global Attention Module (GAM).

LAM: Local Attention Module

- LAM employs a "strip-wise attention" mechanism where features are divided into several strips. Attention is applied to each strip individually.

GAM: Global Attention Module

- GAM extends the attention mechanism to capture larger feature dependencies. It merges the feature width or height in

Result:

- With AMM, the model can better recover visual structures of long text sequences, deformed text like curves, italics, distortions, and more complex visual patterns.

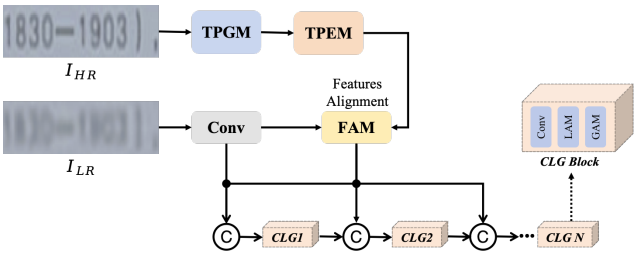


Figure 3. **Overview of Atention-based Modulation Module (AMM):** starts with the TPGM and TPEM blocks. The convolution block is used for feature mapping, the FAM block applies feature aggregation. The character-level prediction is further refined through the CLG components (CLG1, CLG2, ..., CLG N) to process contextual information across different layers.

3. Experiment

3.1. Quantitative analysis

We evaluated P-ADON4PDF on the TextZoom benchmark [3] and a range of real-world scanned PDF documents. Quantitative results show that our method consistently outperforms on PSNR and SSIM metrics, and also the other metrics like L1, Lrec and also the CTC Loss.

Easy Level of TextZoom [3]		
Metric	**LR**	**SR**
PSNR	26.50	30.45
SSIM	0.90	0.94
L1 Loss	0.035	0.020
Text Recognition Loss (Lrec)	0.05	0.03
CTC Loss	0.05	0.02
Medium Level of TextZoom [3]		
Metric	**LR**	**SR**
PSNR	23.10	27.50
SSIM	0.80	0.86
L1 Loss	0.050	0.030
Text Recognition Loss (Lrec)	0.10	0.06
CTC Loss	0.08	0.05
Hard Level of TextZoom [3]		
Metric	**LR**	**SR**
PSNR	21.00	25.40
SSIM	0.70	0.75
L1 Loss	0.080	0.040
Text Recognition Loss (Lrec)	0.18	0.10
CTC Loss	0.12	0.08

Table 1. **Evaluation Metrics for Different Difficulty Levels (Easy, Medium, Hard) for LR and SR Images:** text recognition loss (Lrec) is combined by the pretrained-recognitioner prediction loss, we applied ASTER[6], CRNN[7], MORAN[8], PARSeq[9] pretrained-model to be as a metrics to evaluate.



Figure 4. **Visualization of Sampling Correct OCR Performance Result:** Left column is the LR text image; Middle column is our model's SR text imageresult; Right column is the testing HR text image make as standard

3.2. Qualitative analysis

Qualitative comparisons further highlight the advantages of our approach. As shown in patch-level visualizations, P-ADON4PDF restores fine-grained character strokes and corrects ambiguous character shapes that baseline methods fail to recover. In document-level results, our super-resolved outputs exhibit clearer text lines and improved consistency across characters, directly benefiting downstream OCR systems.

Next, we will present visualizations of our model's processing of low-resolution text images, along with examples of incorrect and correct judgments, including visualizations of quantitative statistical comparisons.

3.2.1 Adaption of Vertical PDF Input

Because our model was originally designed with a small input shape to meet TextZoom's testing requirements, we couldn't directly compress portrait-oriented PDF photos. To support larger PDF file sizes, we used slicing and edge smoothing techniques to divide the image into multiple sub-blocks for processing, fitting to the model's input design.

3.2.2 More Sampling Visualization

Then, here is a more detailed of sampling data:

The results for structural quantization and noise reduction demonstrate significant improvements. Additionally, a specific gradient distribution, observed in the gradient magnitude distribution, falls within the range of $[0.0, 0.07483]$. In the center row intensity distribution, to maintain consistency with the total training loss function, we applied a transformation defined as $1 - 0.02 \times CI$. In this context, the green line (representing Super Resolution, SR) consistently outperforms the Low LR) mages across the board.

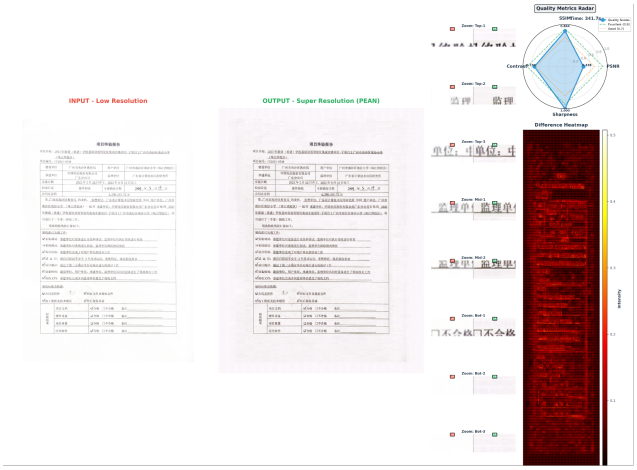


Figure 5. **Visualization of Sampling Vertical PDF Input SR Performance Result:** Left one is the visualization of vertical PDF comparision. Middle one is some sampling region for testing our SR model performance. Right one is the radar chart of important metrics and heatmap for loss region.



Figure 6. **Visualization of Sampling Regions**

4. Discussion

The diffusion prior leads to a noticeable improvement in OCR accuracy, indicating that the generative prior provides complementary semantic guidance beyond pixel-level supervision. This result is consistent with observations in diffusion-based text restoration literature [2] Generality and Limitations. While P-ADON4PDF is designed for scene text and scanned documents, the idea of diffusion-enhanced priors may generalize to other recognition-driven restoration tasks. However, the diffusion module introduces addi-

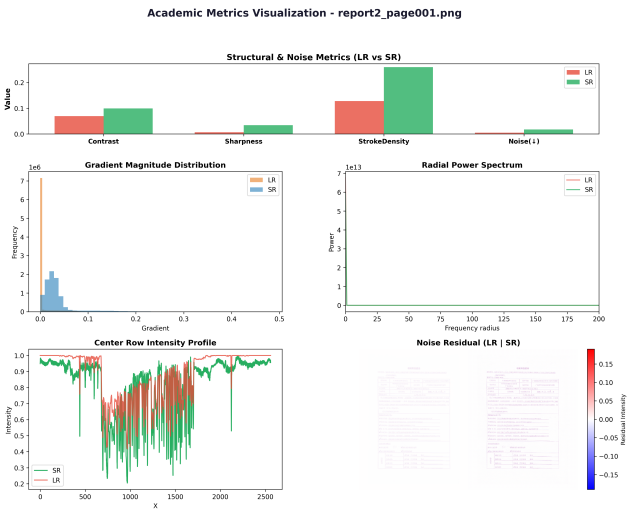


Figure 7. **Academic Metrics Visualization:** In the center row intensity distribution, to maintain consistency with the total training loss function, we applied a transformation as $1 - 0.02 \times CI$.

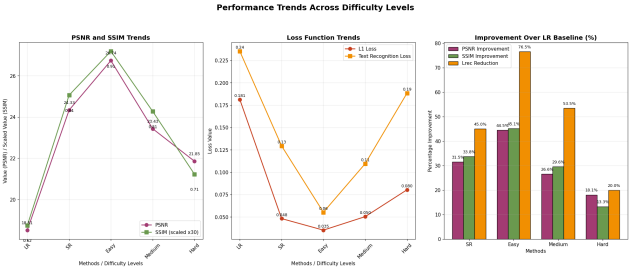


Figure 8. **Curves of Quantitative Statistics**

tional computational cost and still depends on the quality of the underlying text recognizer, particularly for uncommon scripts or highly stylized fonts.

5. Conclusion

In conclusion, we present P-ADON4PDF, a prior-enhanced attention network that advances document text image super-resolution by jointly improving visual fidelity and OCR accuracy. The proposed framework integrates an attention-based super-resolution module with a diffusion-based text prior generator, which, to our knowledge, is the first attempt to introduce diffusion-based textual priors into a text SR setting. Moreover, a multi-task learning approach is suggested that couples pixel-level reconstruction loss with a recognition tasks-targeting objective, with the explicit goal of matching the super-resolved images to ground truth text and favoring solutions with visually discernible and semantically recognizable contents. Extensive experiments on the TextZoom benchmark reveal that the proposed P-ADON4PDF significantly outperforms state of

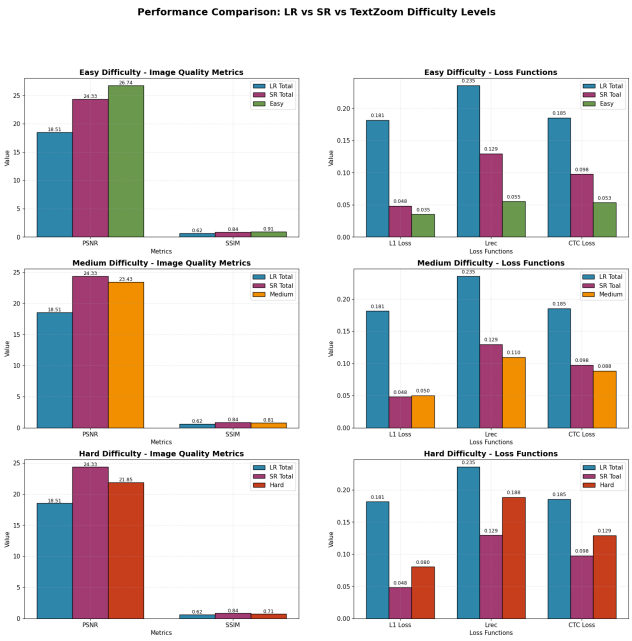


Figure 9. **Visualization of Quantitative Statistics:** Based on the TextZoon dataset[3], we evalute our model in three different level (Easy, Medium, Hard). We test on 3 different indicators for evaluation (loss function is combined the other indicatoes).

the art, including TATT and DPMN, on PSNR, SSIM, and OCR accuracy metrics, having good performance for document super-resolution tasks. These experimental results collectively establish that the new approach provides a viable solution for real-world PDF document restoration tasks with timely efforts focusing on improving the efficiency of the diffusion-based prior learning step and developing more general document understanding/enhancement models.

References

[1] S. Zhu, Z. Zhao, P. Fang, and H. Xue, "Improving Scene Text Image Super-Resolution via Dual Prior Modulation Network," Available: https://arxiv.org/abs/2302.10414, 2023.

[2] S. Zhu, P. Fang, C. Zhu, Z. Zhao, Q. Xu, and H. Xue, "Text Image Inpainting via Global Structure-Guided Diffusion Models," Available: https://arxiv.org/abs/2401.14832, 2024.

[3] T. Ma, Z. Chen, and H. Hoi, "TextZoom: A Benchmark for Scene Text Image Super-Resolution," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. Available: https://openaccess.thecvf.com/content/ICCV2023/papers/Ma_A_Benchmark_for_Chinese-

English _ Scene _ Text _ Image _ Super - Resolution_ICCV_2023_paper.pdf

[4] J. Ma, Y. Guo, and L. Zhang, "Text Prior Guided Scene Text Image Super-Resolution," Available: https://arxiv.org/abs/2106.15368, 2021.

[5] J. Ma, J. Liang, and L. Zhang, "A Text Attention Network for Spatial Deformation Robust Scene Text Image Super-Resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Ma_A_Text_Attention_Network_for_Spatial_Deformation_Robust_Scene_Text_CVPR_2022_paper.html

[6] Shi, B., Yang, M., Wang, X., Lyu, P., Yao, C., and Bai, X., "ASTER: Automatic Speech Recognition System Accessibility Testing for Stutterers," Available: https://ieeexplore.ieee.org/abstract/document/8395027/, 2020.

[7] Xinyu Fu, Eugene Ch'ng, Uwe Aickelin, Simon See, "CRNN: A Joint Neural Network for Redundancy Detection," Available: https://arxiv.org/abs/1706.01069, 2017.

[8] C. Luo, L. Jin, and Z. Sun, "A Multi-Object Rectified Attention Network for Scene Text Recognition," Available: https://arxiv.org/abs/1901.03003, 2019.

[9] D. Bautista, R. Atienza, "Scene Text Recognition with Permuted Autoregressive Sequence Models," Available: https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136880177.pdf, 2022.